Free balancing equations worksheet worksheet 13

Maths | 10th grade | Algebra

1. If 3x + 5 = 2(x + 8) + 1, what is the value of x?

A) x = 3
B) x = 4
C) x = 5
D) x = 6

2. Solve for y: 4(y - 3) = 2y + 6

A) y = 6
B) y = 7
C) y = 9
D) y = 12

3. What is the solution to the equation 2x^2 - 8x + 6 = 0?

A) x = 1 or x = 3
B) x = 2 or x = 3
C) x = 1 or x = 2
D) x = 0 or x = 6

4. If f(x) = 2x + 3, what is f^-1(x)?

A) f^-1(x) = (x - 3)/2
B) f^-1(x) = (x + 3)/2
C) f^-1(x) = 2x - 3
D) f^-1(x) = 2x + 3

5. Simplify the expression: 5x - 3(2x - 4) + x

A) x + 12
B) -x + 12
C) 2x + 12
D) x - 12

6. What is the solution set of the inequality 3x - 4 > 2(x + 1)?

A) x > 2
B) x > 0
C) x < 2
D) x < 0

7. Given that g(x) = x^2 - 5x + 6, find g(3).

A) 3
B) 0
C) 6
D) 9

8. If h(x) = 3x - 7 and j(x) = -x + 2, what is (h " j)(x)?

A) 3x + 1
B) -3x + 11

C) 3x - 11
D) -3x + 14

9. Which of the following is the vertex of the parabola y = x^2 - 4x + 3?

A) (2, -1)
B) (2, 3)
C) (1, -2)
D) (3, 2)

10. Determine the nature of the roots of the equation x^2 + 4x + 5 = 0.

A) Two real roots
B) Two distinct real roots
C) Two complex roots
D) One real root

Answer Key

1. A
Solving: 3x + 5 = 2x + 16 + 1 => 3x + 5 = 2x + 17 => x = 12

2. C
Expanding: 4y - 12 = 2y + 6 => 4y - 2y = 6 + 12 => 2y = 18 => y = 9

3. A
Using quadratic formula: x = [8 ± sqrt((-8)^2 - 4(2)(6))]/(2*2) = 1, 3

4. A
Swap x and y: x = 2y + 3 => y = (x - 3)/2

5. C
Expanding: 5x - 6x + 12 + x = 2x + 12

6. B
3x - 4 > 2x + 2 => x > 6

7. B
Substitute x = 3: g(3) = 3^2 - 5(3) + 6 = 0

8. B
(h " j)(x) = h(j(x)) = h(-x + 2) = 3(-x + 2) - 7 = -3x + 6 - 7 = -3x + 11

9. A
Vertex form: y = (x - 2)^2 - 1, vertex is (2, -1)

10. C
Discriminant: b^2 - 4ac = 16 - 20 = -4; negative implies complex roots